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Honeycomb carbon fibers strengthened composite phase change materials for superior thermal energy storage

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HIGHLIGHTS

- Thermal conductive property of paraffin was enhanced by honeycomb carbon fibers.
- Honeycomb carbon fiber bundles were prepared from biomass sisal.
- The composite PCM presented a high thermal conductivity of 1.77 W m^{-1} k^{-1}.
- The composite PCM showed an anisotropic and high thermal conductivity along the fiber direction.
- The composite exhibit good form-stability and high thermal capacity.

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ABSTRACT

Phase change materials (PCMs) have shown promising applications for thermal energy storage and management. With the purposes of solving the critical leakage problem and improving the thermal conductive property of paraffin PCM, composite PCMs as-supported by carbon fiber bundles were fabricated by a simple vacuum impregnation. The porous and honeycomb-shaped carbon fibers were prepared by the direct carbonization of biomass sisal fibers. As beneficial from the carbon supporting scaffolds, the composites illustrated great thermal conductivity increasement (1.73 W m^{-1} K^{-1}) over pure paraffin (0.25 W m^{-1} K^{-1}). The composite PCMs show satisfying latent heat (192.2 J g^{-1} at a carbon ratio of 12.8%), good shape stability, favorable thermal reliability and cyclability. Moreover, due to the one-dimensionally arranged carbon fiber bundles, the composites also showed anisotropic thermal conductive property. All of those could reinforce the use of the low-cost sisal-derived carbon fibers as thermal conductive scaffolds to PCMs, which will promote their practical applications for advanced thermal energy storage in collecting solar energy and industrial waste heat.

1. Introduction

Phase change materials (PCMs) have shown many potential applications in the thermal energy storage and management fields, such as solar energy collection, waste heat recovery, energy-saving building and thermal management of electronic devices. Thermal energy can be stored/released during the phase transition of a PCM, such as the typical solid-liquid phase change. [1–4] In order to satisfy the requirements of the practical use for thermal energy storage and management, a PCM should have several features, such as high latent heat, high thermal conductivity, good chemical stability, nontoxicity, easy availability and good form-stability. Under this circumstance, organic PCMs including polyethylene glycol [5,6], paraffin wax (PW) [7,8] and fatty acids [9] are good candidates. However, the common disadvantages of organic PCMs, including the poor thermal conductivity and easy leakage in the liquid state, have prevent their widespread use. Here, for the practical application, a high thermal conductivity is required which determines the efficiency of thermal energy collection and releasing. On the other hand, an excellent encapsulation capability against leakage can ensure the long-term service of the system.

A lot of efforts have been dedicated to preparing composite PCMs with increased thermal conductivity by adding thermal conductive fillers, such as metallic materials, ceramic substances and carbon materials including graphite, porous carbon, graphene and carbon

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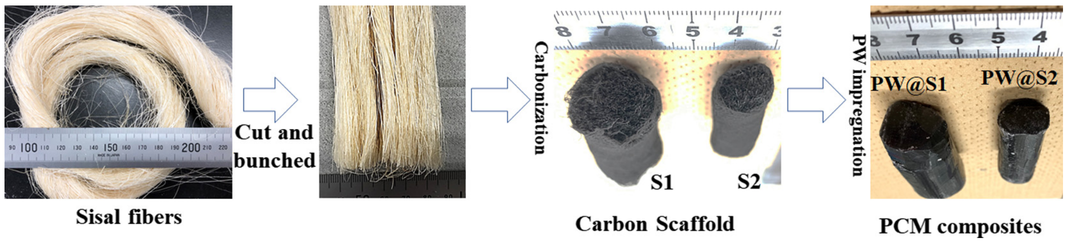

Fig. 1. Schematic diagram for the preparation of sisal-derived carbon scaffolds and PCM composites.

nanotubes (CNTs). [9–16] The addition of powdery fillers that are dispersed in the PCM matrix can somewhat improve the thermal conductivity, however, the powders are not stable when PCM is in the liquid state. In addition, to obtain a satisfied thermal conductivity, a large amount of filler addition is needed, which could unavoidably decrease the thermal capacity of the composites. The fabrication of pre-constructed three-dimensional (3D) porous frameworks of the thermal conductive fillers is a good choice. The 3D form-stable porous framework can offer continuous thermal conductive network. Additionally, the porous structure is also important for the incorporation of PCM which could prevent the liquid PCM from leakage. In such attempts, 3D porous supports such as graphite foams [17,18], graphene aerogels [19,20], CNT scaffolds [21] and BN foams [22] have been used by several research groups. Among them, the use of high-quality porous carbon supports such as graphene or CNT frameworks have shown largely enhanced thermal transfer properties of composite PCMs. However, the high-cost and complex production of these carbon supports are the drawbacks for their broad application.

Biomass is the non-expensive, abundant and sustainable resources for producing carbon. Researchers have tried to produce porous carbon from biomass, such as cotton cellulose [12,23], starch [24], lignin [25], wood [7] and so on. These porous carbons have been investigated for a variety of applications, such as for the electrode active materials in Li ion batteries [23], supercapacitors [24], and as the conductive supporting materials for PCMs [7,12,25]. As for the thermal conductive additives to PCMs, Yang H. et al. reported the use of carbonized wood as thermal conductive supporting scaffold to PW, and an enhanced thermal conductivity of 0.669 $W\,m^{-1}\,K^{-1}$ is obtained [7]. Atinafu D. et al reported the preparation of porous carbon from cotton cellulose using a $Mg(OH)_2$ template method, and the porous carbon products were used as fillers to hexadecanol. An increased thermal conductivity of 0.41 $W\,m^{-1}\,K^{-1}$ for the composites with a carbon content of around 15% can be achieved as compared to hexadecanol with a value of 0.35 $W\,m^{-1}\,K^{-1}$ [12]. However, the thermal conductivity improvement is still limited. Therefore, it is still of great interest to use biomass carbon for thermal conductive support of PCMs, especially by maximizing the structure characteristics of the biomass carbon with special structures, such as one-dimensional (1D) fibers.

It is known that the enhancement of the thermal conductivity can be affected by the geometry and alignment patterns of the fillers, especially when the high aspect ratio materials are used, such as 1D fibers and 2D sheets or flakes. For 1D fibrous fillers, if the fibers can be arranged along the unique anisotropic fiber direction, anisotropic thermal conductive network can be achieved. This is good for the maximum utilization of the 1D structural characteristics. By employing the aligned fiber scaffold for supporting PCMs, the composites with anisotropic thermal conductivity are especially useful for unique applications such as fast thermal dissipation of battery packages and electronic devices, and solar thermoelectric generator.

In this study, the honeycomb structured carbon fibers are produced by the simple carbonization of biomass sisal fibers. By pre-aligning the sisal fibers, carbon fiber bundles with anisotropic thermal conductivity can be obtained. Paraffin is impregnated into the carbon scaffolds to fabricate form-stable composite PCMs. The thermal storage and heat transfer properties of the composite PCMs are carefully investigated.

2. Experiment

2.1. Preparation of sisal-derived carbon scaffolds and form-stabilized composite PCMs

Sisal fibers that could be longer than 1 m were used as the raw materials. In a controlled experiment, the sisal fiber bundles were cut to a length of 5 cm. The fiber bundles were well fixed by wrapping with cotton gauze. The sisal fiber bundles were pyrolyzed and carbonized under Ar atmosphere up to 2400 °C. In order to prepare carbon scaffolds with different densities, the same amount of sisal fibers were closely bunched to cylinders with different diameters of around 2.4 cm and 1.8 cm. After carbonization, two cylinder-shaped carbon scaffolds with diameters of around 2 cm and 1.5 cm were obtained, which were named as S1 and S2. The two carbon scaffolds have density of around 0.097 and 0.172 $g\,cm^{-3}$, respectively. PW ($C_{21}H_{27}NO_3$) provided by Kishida Chemical Co., Ltd. was used as the PCM. The PCM composites were prepared by vacuum impregnation of PW into the carbon scaffolds. The composite PCMs were named as PW@S1 and PW@S2, respectively.

The preparation procedures and optical pictures of the samples at each step for the carbon scaffolds and composite PCMs are shown in Fig. 1.

2.2. Characterization

Powder X-ray diffraction (XRD) patterns of the samples were measured by an X-ray diffractometer (Rigaku, Cu-Kα). The microstructure of carbon and PCM composites were investigated by scanning electron microscope (ZEISS, Sigma-500). The specific surface area and pore size distribution of the carbon were measured by nitrogen absorption (BELSORP-mini). The specific surface area was calculated by Brunauer-Emmett-Teller (BET) method, while the pore size distribution was analyzed by Barrett-Joyner-Halenda (BJH) method. Thermal gravimetric (TG, Netzsch, STA 2500 Regulus) analysis under air flow were conducted for the biomass carbon, PW and PCM composites at a heating ramp of 10 $°C\,min^{-1}$. The thermal conductivity of the composites and PW was measured by a thermal conductivity analyzer. The thermal properties including phase change enthalpies and phase change temperatures were analyzed by a differential scanning calorimeter (DSC). The leakage test of the samples was examined using the digital camera method. In addition, the heat transfer performance in both heating and cooling processes were also measured by an infrared camera. The Fourier transform infrared (FT/IR) spectra of the samples before/after cycling were obtained by a FT/IR 660Plus spectrometer.

3. Results and discussion

3.1. Preparation and characterization of sisal-derived carbon scaffolds and composite PCMs

Fig. 1 shows the preparation procedures for the carbon scaffolds and composite PCMs. Sisal fiber bundles with aligned fibers were used as

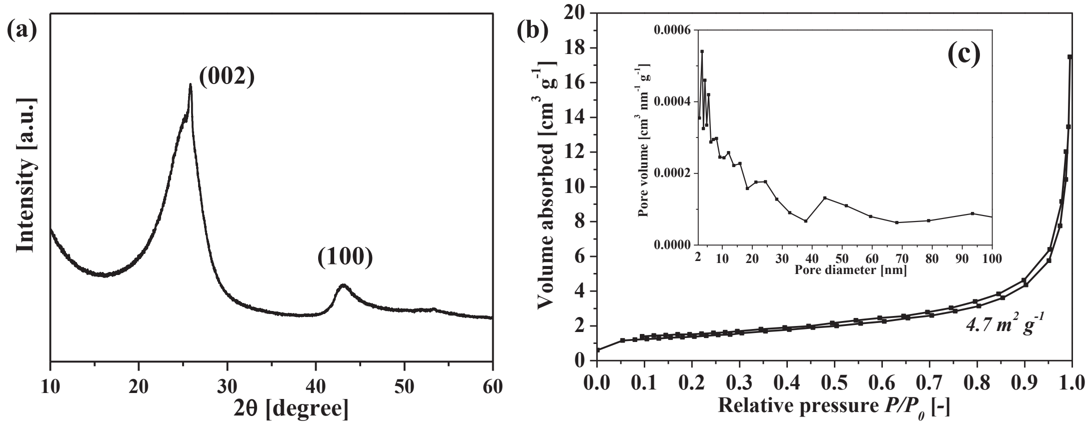

Fig. 2. XRD pattern (a), nitrogen adsorption isotherm (b) and pore size distribution (c) for the sisal-derived carbon at 2400 °C.

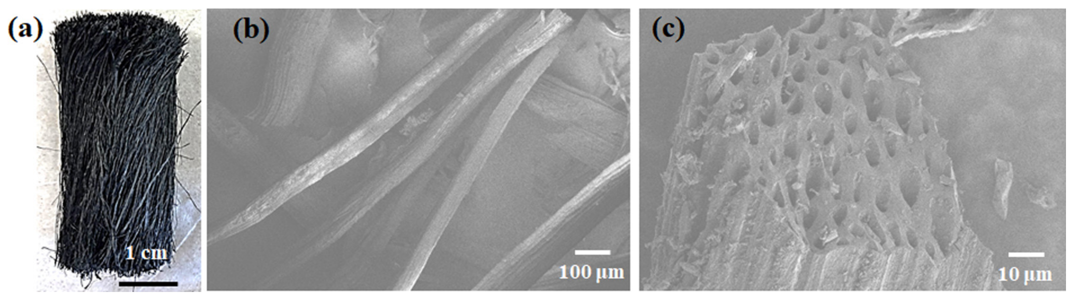

Fig. 3. Morphology observation of sisal-derived carbon. (a) Photo picture; (b, c) SEM images.

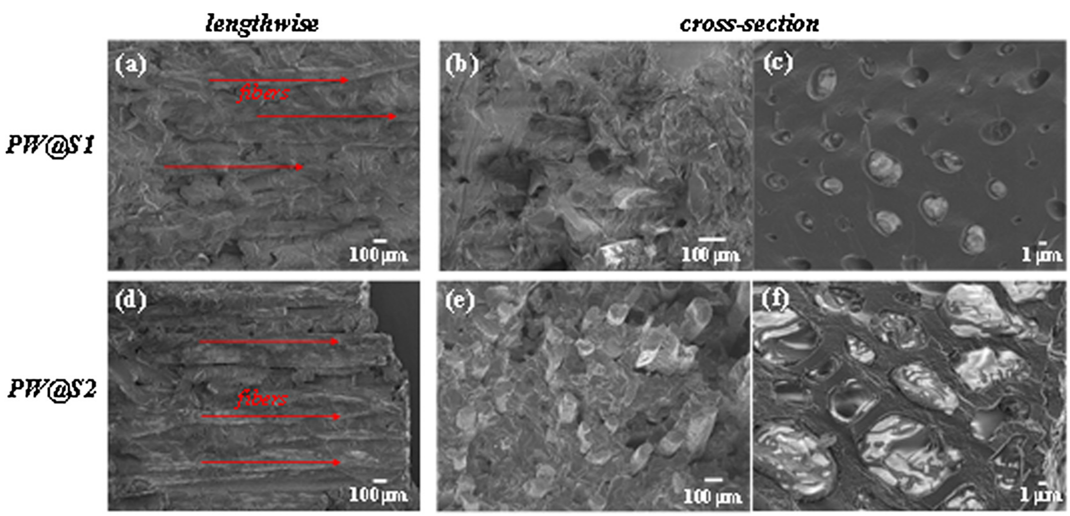

Fig. 4. SEM images of the PCM/carbon composites observed from the lengthwise and cross-section of the fibers.

the raw materials, and after a simple carbonization process, carbon scaffolds consisted of aligned carbon fibers were fabricated.

The XRD pattern of the carbonized sisal at 2400 °C is shown in Fig. 2-(a). Two broad peaks corresponding to the planes of carbon (0 0 2) and (1 0 0) are observed. Fig. 2-(b) shows the nitrogen adsorption isotherm of the carbon, while Fig. 2-(c) presents the BJH pore size distribution. The BET specific surface area is 4.7 $m^2\ g^{-1}$ for the carbon sample. The carbonized sisal fibers were further observed by optical camera and SEM, as shown in Fig. 3. The long and black carbon fibers were obtained after carbonization. The original sisal fibers can be as long as to several tens of centimeters, and the fibers have diameters of around 100 µm. After carbonization, the fibers can retain the original shape with diameters of around 100 µm. It is also noted that the fibers are not dense but contain many honeycomb-like holes in the size of several micrometers, as shown by the cross-section image of the fibers. This is a very good characteristic of the sisal-derived carbon, since that the holes can be effectively used for encapsulating PCMs.

The scaffolds of sisal-derived carbon fibers were used as the supporting frameworks for organic PCM. By vacuum impregnation, liquid PW were incorporated into the holes of the fibers and the porous spaces of the fiber bundles. Fig. 4 shows the SEM images of the composites observed from the lengthwise and cross-section of the composite cylinders. From the SEM images observed from the lengthwise direction, we can see that the sisal-derived fibers are aligned in the same direction as incorporated in the PW matrix. From the cross-section SEM images, it is observed that the holes of the carbon fibers and the porous spaces of the carbon scaffolds are filled with PW. It is also observed that PW@S2 presents higher density of carbon fibers as the filler than that of PW@

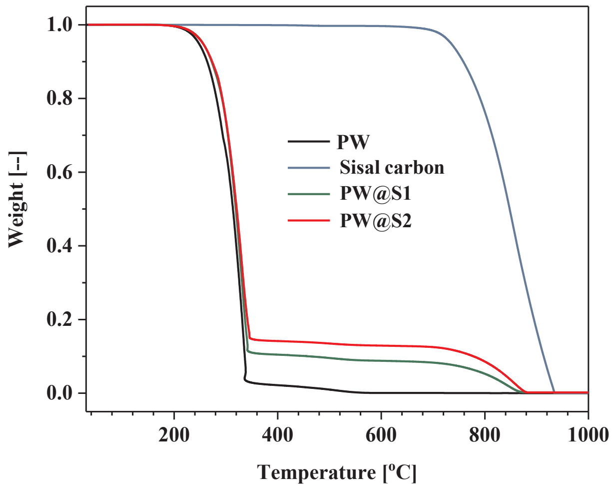

Fig. 5. TG curves for the air combustion of PW, sisal carbon and PCM composites.

S1.

TG analysis under air flow was also used to evaluate the thermal stability of the sisal-derived carbon. As shown in Fig. 5, it is seen that the air-combustion of sisal-derived carbon starts at around 700 °C, indicating its good thermal resistance property. The air-combustion behavior of the PW and PCM composites were also analyzed by TG, as compared in Fig. 5. The combustion of PW starts at around 250 °C and finishes at around 350 °C. However, the combustion of the PCM composites shows two stages of weight losses in the TG curves. The one in the temperature range of ~250–350 °C is due to the combustion of PW, while the one in the high temperature side of above 700 °C is due to the combustion of sisal carbon. By comparing the weight changes in these two stages, the weight ratios of PW in the composites can be calculated. The PW weight ratios are 91.3% and 87.2% for PW@S1 and PW@S2, respectively. The ratios are similar to those as calculated by comparing the weight changes before/after PW impregnation.

3.2. Phase change properties of the composites and PW

The phase change and thermal properties of the composites and PW were investigated by DSC measurement. Fig. 6 and Table 1 present the results obtained from DSC analysis. The shapes of DSC curves and the phase change temperatures of PW and PCM composites are quite similar, indicating that the porous carbon scaffold does not influence solid-liquid phase transition of paraffin. Two peaks in both the heating and cooling processes are observed for PW and composites. The weak peaks in the low temperature side are ascribed to the solid-solid phase transition, while the strong peaks in the high temperature side represent the solid-liquid phase transition. During the 100 cycles of melting-solidification, all samples present very good reversibility, which is indicated by the almost overlapped DSC curves. The enthalpies during the 100 cycles are also very stable as shown both in Fig. 6-(d) and Table 1. These results illustrate that PW is a good thermal storage media that can be used for long-term repeated use. PW has a latent heat capacity of ~217.7 J g^{-1}. For the composite PCMs, the latent heat capacities are 201.6 and 192.2 J g^{-1}. The composites have lower heat capacities compared to PW due to the increased weight of carbon filler.

3.3. Chemical stability analysis

The chemical stability of the composite PCM was checked by FR/IR

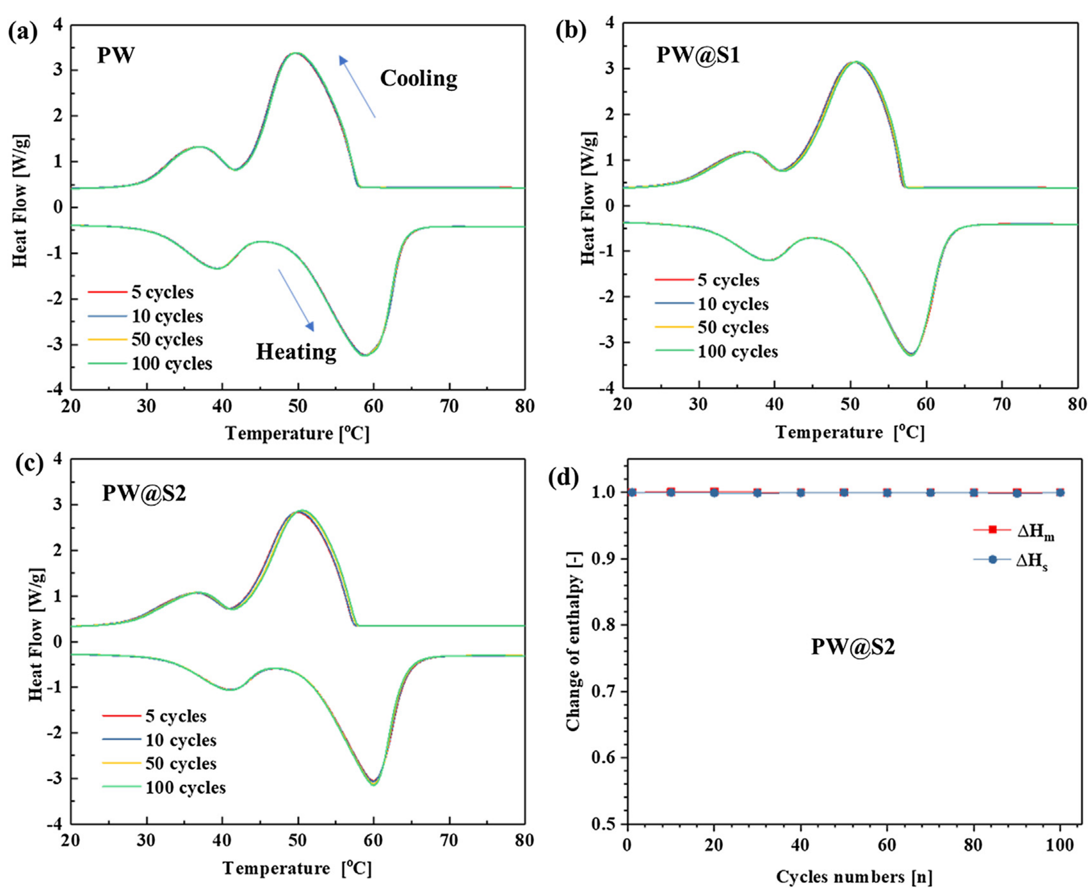

Fig. 6. DSC analysis results PW and PCM composites.

Table 1
Summarized DSC parameters of the PCM composites and PW.

Samples	Cycle times	*Melting process*				*Solidification process*			
		T_{mo} (°C)	T_{me} (°C)	T_{mp} (°C)	ΔH_m (J g^{-1})	T_{so} (°C)	T_{se} (°C)	T_{sp} (°C)	ΔH_s (J g^{-1})
PW	5	25.5	66.0	57.8	217.7	25.0	58.4	50.5	215.4
	100	25.5	66.0	57.8	217.7	25.0	58.4	50.5	215.3
PW@S1	5	22.1	67.9	58.0	201.6	22.4	57.1	50.4	200.9
	100	22.1	67.9	57.9	201.3	22.5	57.1	50.7	200.8
PW@S2	5	22.5	69.2	60.0	192.2	22.5	57.6	50.0	191.1
	100	22.5	69.1	60.0	192.1	22.5	57.7	50.5	190.9

T_{mo}, T_{me}, T_{mp}, ΔH_m are the onset point, end point, peak temperature and phase change enthalpy during the melting process, while T_{so}, T_{se}, T_{sp}, ΔH_s are the onset point, end point, peak temperature and phase change enthalpy during the solidification process, respectively.

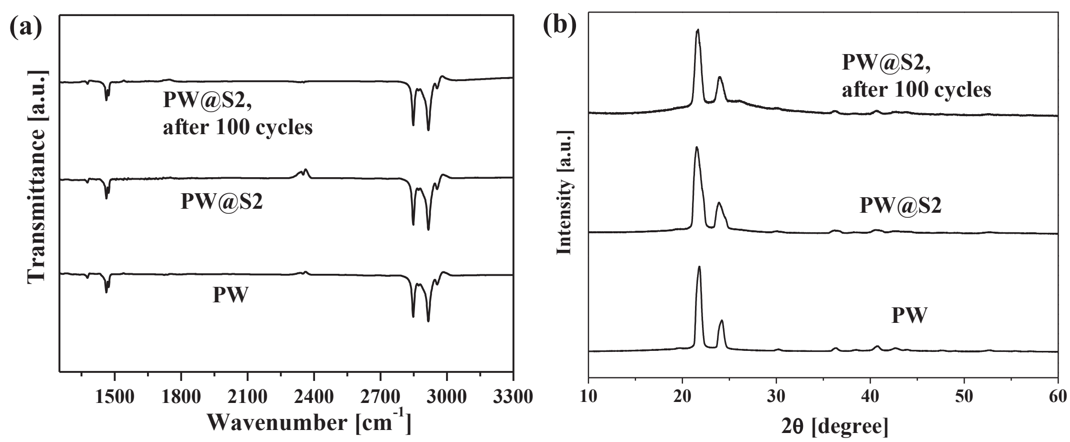

Fig. 7. FT/IR spectra (a) and XRD patterns (b) for the composite before/after cycling in comparison with PW.

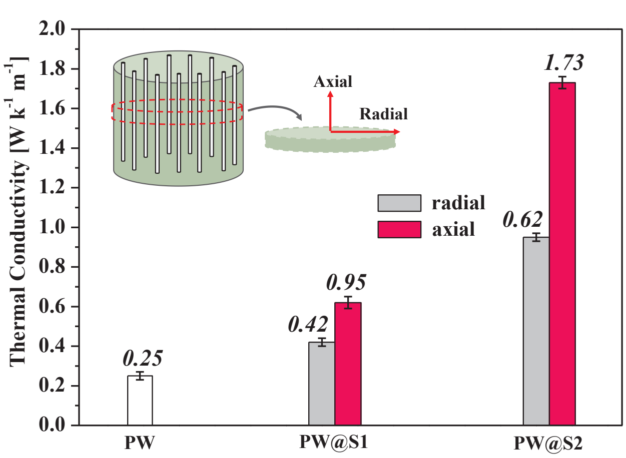

Fig. 8. Thermal conductivity of PW and the composites.

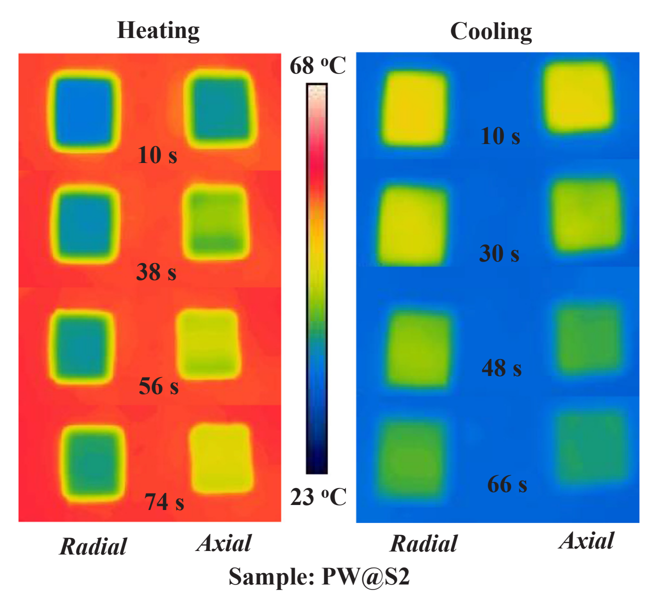

Fig. 9. Thermal images obtained by an infrared camera during both the heating and cooling processes for the composite observed from the Radial and axial directions.

and XRD analysis. Fig. 7 shows the FR/IR spectra and XRD patterns of one composite before/after cycling in DSC test for 100 cycles, in comparison to original PW. The peaks for these samples are identical for both FT/IR and XRD analysis, indicating the good stability of PW. Additionally, the sisal-derived carbon at 2400 °C is quite stable, which does not react with PW. Therefore, the good chemical and thermal stability of both PW and carbon supporting material could reinforce their long-term use for thermal energy storage.

3.4. Thermal conductive properties

The thermal conductivity of the samples was measured by thermal conductive analyzer at room temperature, and the apparent heat transfer characteristics were confirmed an infrared camera. The thermal conductivity of the composite PCMs was measured from the axial and radial direction of the carbon fibers, in which the samples were cut and shaped to disks from these two measurement directions. The thermal conductivity of PW is very low with a value of ~0.25 W m^{-1} K^{-1}. With the incorporation of carbon fibers, the thermal conductivity of the composites can be greatly enhanced, especially from the axial direction of the fibers. As shown in Fig. 8, the increasement of thermal conductivity form the radial direction is limited, and the values

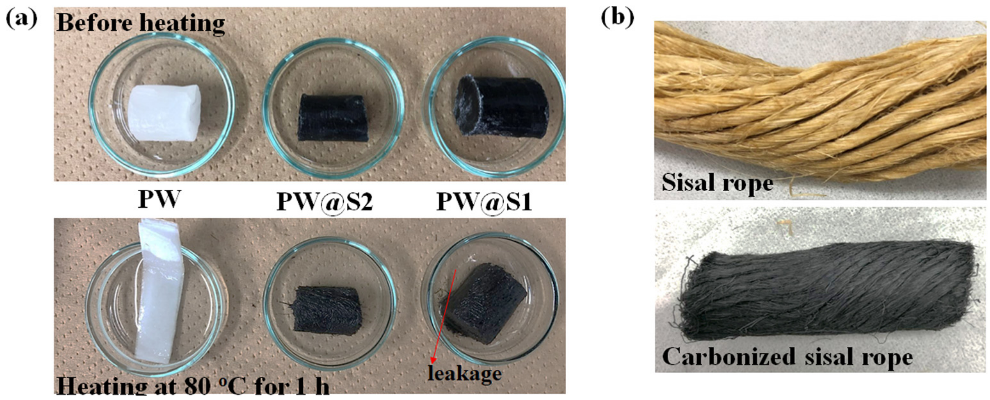

Fig. 10. Shape-stability analysis. (a) Leakage test of PW and the composites; (b) a consolidated shape design by forming sisal rope.

are 0.42 and 0.62 $W\,m^{-1}\,K^{-1}$ for PW@S1 and PW@S2, respectively. However, the corresponding values from the axial direction are as high as 0.95 and 1.73 $W\,m^{-1}\,K^{-1}$, respectively. The very long fibers can offer continuous thermal conductive path along the axial direction of the carbon fibers. In comparison, in the radial direction, a well-connected heat pass way is not efficiently formed, which induces limited increasement of thermal conductivity in the radial direction even under a high filler addition content. In fact, the carbon fibers can be somewhat separated by PW, as shown by the SEM images in Fig. 4. Therefore, the PCM composites filled with aligned carbon fibers present anisotropic thermal conductivity.

In order to further confirm the anisotropic thermally conductive property of the composite filled with sisal carbon fibers, the heat transfer features in both the heating and cooling processes from the axial and radial directions are compared by using an infrared camera. As for the heating process, the PCM composite blocks were put on ceramic substrate which was heated at 80 °C, and the thermal pictures at different durations were recorded. For the cooling process, the composite blocks were firstly heated at 80 °C for the same duration, then they were transferred to a cool ceramic substrate at room temperature, and the thermal images were also recorded. The results are shown in Fig. 9. It is obvious that for the composite block observed from the axial direction, the sample present higher heat transfer speed both in the heating and cooling processes. As a summary, by arranging the fibers in the same direction, an anisotropic thermal conductive network was successfully achieved. The 1D fibers can maximize their structural features, promoting the greatly enhanced thermal conductive property along the fiber direction. Such anisotropic thermally conductive composites are especially useful for unique applications such as fast thermal dissipation of battery packages and electronic devices, and solar thermoelectric generator.

3.5. Leakage test and the shape design of the carbon scaffold

Shape-stability is a crucial parameter for the practical use of organic PCMs. The leakage test was carried out by heating the composites and PW at 80 °C and the shape-stability was investigated by digital camera observation, as shown in Fig. 10-(a). After heating at 80 °C for 1 h, PW can be totally melted which spreads over the container. As for the composites, PW@S1 presents a little leakage since that the carbon scaffold is loose which cannot absorb all of the liquid PW. However, for composite PW@S2, because of the dense carbon scaffold, the leakage of liquefied PW can be prevented. Here, the denser carbon network of S2 with proper porous structure can offer enough absorption force to confine liquid PW due to the capillary effect. Additionally, as a consolidated shape design for the sisal-derived carbon scaffold, it can be twisted to yarns and ropes, as shown in Fig. 10-(b).

4. Conclusion

In conclusion, composite PCMs with enhanced thermal storage performances were prepared by vacuum incorporation of paraffin into porous carbon scaffolds that were consisted of carbon fiber bundles. The porous and honeycomb-shaped carbon fibers were prepared by the direct carbonization of biomass sisal fibers. By pre-aligning the sisal fibers, carbon fiber bundles with anisotropic thermal conductive property were obtained. As beneficial from the carbon supporting scaffolds, the composite PCMs not only illustrated a high thermal conductivity (1.73 $W\,m^{-1}\,K^{-1}$) at a carbon filler content of 12.8%, but also resulted in good shape stability. The composites showed good thermal stability and reversibility with almost 100% retention after 100 cycles of melting-solidification. Finally, the biomass raw material was inexpensive, and the preparation of sisal-derived carbon scaffold was very simple. In addition, the fibers could be easily shaped to diverse architectures for various applications. All of those could promote the practical applications of sisal-derived carbon supported composite PCMs for thermal energy storage.

Declaration of Competing Interest

The authors declared that there is no Conflict of Interest.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.applthermaleng.2019.114493.

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